



Moisture Measuring



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Compiled By: D. Losty		Moisture Measuring	Revision 10
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1 Overview

- 1 Prior to treatment you must check all poles for moisture content to ensure that they have a maximum moisture content of:
 - Poles to be creosote treated - $\leq 28\%$. Where any obtained reading is above 25% you must ensure that the moisture content is confirmed by oven testing.
 - Pole to be treated with water-borne preservatives – $\leq 25\%$.
- 2 Where any poles exceed the above maximum values, you may still fabricate the poles, with the following additional requirements:
 - You must ensure that fabrication of the poles in a 'wet state' has been authorised by the Production Supervisor.
 - Once fabrication has been completed, you must securely attach a blue label that contains a large letter 'F' (to indicate that the pole has been fabricated in a 'wet state'), together with the month and year that the moisture reading was obtained (see example label in Module 1, table 1.2)

2 Testing Locations

2.1 Locations at Which Moisture Testing Must be Conducted

Individual moisture readings must be obtained at the locations shown in the following table and dependent upon the customer for whom the pole is to be fabricated:

Customer	Testing Locations
BT	1x reading at 1.0m, 1.5m and 2.5m from the butt of the pole and at 120° spacing around the pole.
Scottish Power	2x readings (spaced 180° apart) at 3.0m from the butt of the pole and 2x readings (spaced 180° apart) at the mid-point of the pole.
All other RECs	2x readings (spaced 180° apart) at 0.4m, 1.5m & 3.0m from the butt of the pole.

3 Using a Resistance Pin Meter

3.1 Features

- 1 The Protimeter Timbermaster is a resistance type moisture meter designed for use in wood.

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- Moisture measurements can be taken using the integral pin electrodes, or by using the meter in combination with moisture probes or hammer electrodes. Because of the depth of reading required, you must only use this meter with hammer electrodes.



- This model of moisture meter measures the electrical resistance of the timber between pin electrodes. The resistance constant of the material is then related (via the internal software and taking into account species density) to a moisture level.
- The meter dimensions are 180 x 28 x 49.
- The meter can accurately measure between 7%mc and the fibre saturation point of the wood. The entire measuring range of the meter is 7%mc to 99.9%mc. However, any reading over the fibre saturation point of the wood cannot be deemed to be accurate.
- The maximum read depth of the meter is 32mm.

3.2 Operating Parameters

- The operating temperature range of this meter is from 0°C to 50°C.
- The Timbermaster meter is calibrated for wood at 20°C (70°F). In general, timber that is hotter than 20°C will give higher readings and timber colder than 20°C will give lower readings. For this reason:
 - Where the timber is above 20°C, you must **subtract** approximately 0.5% moisture content per 5°C from the measured value.
 - Where the timber is below 20°C, you must **add** approximately 0.5% moisture content per 5°C to the measured value.
- The meter features eight calibration scales, enabling accurate moisture measurements in 150 wood species.
- There is no minimum thickness of wood that can be accurately measured.

3.3 Pre-Use Verification of Calibration

- At the beginning of each day, you must check that the calibration of the meter is within the required parameters, using the 'CalCheck' tab that is provided for this purpose (or a Protimeter Checkbox).

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2 You must conduct the calibration check as follows:

- Check the condition of all wires and connections. Ensure that there are no breaks in the wires and that all connections are secure.
- Check that both of the knurled clamping wheels on the hammer probe are fully tightened.
- Check that the paint coating on the electrode pins has not worn away and that the only exposed metal is at the pointed tip of each pin.
- Switch on the meter and ensure that the scale is set to 'A'.
- Hold the electrode needles across the exposed wires of the 'calcheck' device (or across the terminals of the Protimeter Checkbox).
- Note the moisture reading displayed.
- If the moisture reading falls within the range of 17.8% to 18.3% then you can use the meter. If this is not the case, you must not use the meter and you must be return it to the Production Supervisor, together with a description of the defect.

3.4 Operating Procedure

- 1 Inspect the meter for signs of damage, contamination or moisture. Pay particular attention to pin electrodes, ensuring that they:
 - Are free from dirt and contaminants (such as resin)
 - Are straight
 - Are firmly secured to the hammer probe body
 - Are coated along their length (except the pointed tips) with powder coat paint that is not worn away
- 2 Securely connect the hammer electrode to the 3.5 mm socket on the right hand side of the meter.
- 3 Switch the meter on by pressing 
- 4 Select the appropriate wood calibration scale (A, B, C, E, F, G, H or J) by referring to the timber calibration tables (A2.6) and press 
- 5 Drive the hammer electrode pins into the wood to their full depth.

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- 6 Note the moisture content reading.
- 7 Normal readings are displayed in solid characters. However, if an obtained reading is above the fibre saturation of the wood, the characters in the display will become faded and grey.
- 8 If the meter or hammer probe are dropped (or otherwise suffer an impact) you must be re-check the calibration using the 'CalCheck' tab.

3.5 Selection of Testing Locations

- 1 The meter calculates moisture content taking into account the density of the wood. For this reason you must avoid the following locations when obtaining measurements:
 - Surface checks (also known as weathering checks).
 - Knots – The varying densities within the knot (heartwood, sapwood and deviated grain) will affect the calculations and return a potentially inaccurate reading.
- 2 The ideal conditions for obtaining accurate measurement are areas that are clear of surface checks, knots, in-barks or felling machine indentations.

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3.6 Timber Calibration Tables



Calibration Tables for Wood

Timber Species Group Table
Common names of timbers as
BS888 & 589: 1973

Abura E Afara A Aformosa G Afzelia E Agba J Amboyna G Ash, American B Ash, European A Ash, Japanese A Ayan C Baguacu, Brazilian F Balsa A Banga Wanga A Basswood G Beech, European C Berlina B Binvang E Birch, European J Birch, Yellow A Bisselon E Bitterwood F Blackbutt C Bosquiea A Boxwood, Maracaibo A Camphorwood, E African C Canarium, African B Cedar, Japanese B Cedar, West Indian J Cedar, Western Red C Cherry, European J Chestnut C Coachwood G Cordia, American Light F Cypress, E African A Cypress, Japanese (8-18%mc) J Cypress, Japanese (18-28%mc) C	Dahoma Danta Douglas Fir Elm, Japanese Grey Bark Elm, English Elm, Rock Elm, White Empress Tree Erimado Fir, Douglas Fir, Grand Fir, Noble Gegu, Nohor Greenheart Guarea, Black Guarea, White Gum, American Red Gum, Saligna Gum, Southern Gum, Spotted Gurjun Hemlock, Western Hiba Hickory Hyedunani Iroko Ironbark Jarrah Jelutong Kapur Karri Kauri, New Zealand Kauri, Queensland Keruing Kuroka Larch, European Larch, Japanese Larch, Western Lime Loliondo Mahogany, African Mahogany, West Indian Makore Mansonina Maple, Pacific Maple, Queensland Maple, Rock Maple, Sugar Matai Meranti, Red (dark/light) Meranti, White Merbau Missanda Muhuhi Muninga	A C B B E E E J F B A J H C J H A B A B A A C J F B F B C C A A E J F A C C F E C J B B B A A E B B B C J G
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Musine	J	Silky Oak, African	C
Musizi	J	Silky Oak, Australian	C
Myrtle, Tasmanian	A	Spruce, Japanese (8-18%mc)	J
Naingon	C	Spruce, Japanese (18-28%mc)	C
Oak, American Red	A	Spruce, Norway (European)	C
Oak, American White	A	Spruce, Sitka	C
Oak, European	A	Stringybark, Messmate	C
Oak, Japanese	A	Stringybark, Yellow	C
Oak, Tasmanian	C	Sterculia, Brown	A
Oak, Turkey	E	Sycamore	F
Obeche	G	Tallowwood	A
Odoko	E	Teak	F
Okwen	B	Totara	E
Olive, E African	B	Turpentine	C
Olivillo	G	Utile	J
Opepe	H	Walnut, African	J
Padang	A	Walnut, American	A
Padauk, African	F	Walnut, European	C
Panga Panga	A	Walnut, New Guinea	B
Persimmon	G	Walnut, Queensland	C
Pillarwood	F	Wawa	G
Pine, American Long Leaf	C	Wandoo	J
Pine, American Pitch	C	Whitewood	C
Pine, Bunya	B	Yew	C
Pine, Caribbean Pitch	C		
Pine, Corsican	C	Botanical Names of Timbers	
Pine, Hoop	C	Abies alba	B
Pine, Huon	B	Abies grandis	A
Pine, Japanese Black	B	Abies procera	J
Pine, Kauri	E	Acanthopanax ricinifolius	A
Pine, Lodgepole	A	Acer macrophyllum	A
Pine, Maritime	B	Acer pseudoplatanus	F
Pine, New Zealand White	B	Acer saccharum	A
Pine, Nicaraguan Pitch	C	Aetoxicon punctatum	G
Pine, Parana	B	Aformosia elata	G
Pine, Ponderosa	C	Azalia spp	E
Pine, Radiata	C	Agathis australis	E
Pine, Red	B	Agathis palmerstoni	J
Pine, Scots	A	Agathis robusta	J
Pine, Sugar	C	Amblygonocarpus andogensis	A
Pine, Yellow	A	Amblygonocarpus obtusungulis	A
Poplar, Black	A	Araucaria angustifolia	B
Pterygota, African	A	Araucaria bidwilli	B
Pyinkado	E	Araucaria cunninghamii	C
Queensland Kauri	J	Berlinia grandiflora	B
Queensland Walnut	C	Berlinia spp	B
Ramin	G	Betula alba	J
Redwood, Baltic (European)	A	Betula alleghaniensis	J
Redwood, Californian	B	Betula pendula	J
Rosewood, Indian	A	Betula spp	J
Rubberwood	H	Bosquiera phoberos	A
Santa Maria	H	Brachylaena hutchinsii	J
Sapele	C	Brachystegia spp	B
Sen	A	Calophyllum brasiliense	H
Seraya, Red	C		

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Canarium schweinfurthii	B	Juglans nigra	A
Cardwellia sublimis	C	Juglans regia	C
Carya glabra	F	Khaya senegalensis	E
Cassipourea elliotii	F	Khaya ivorensis	J
Cassipourea melanosana	F	Larix decidua	C
Castanea sativa	C	Larix kaempferi	C
Cedrela odorata	J	Larix leptolepis	C
Ceratopetalum apetalum	G	Larix occidentalis	F
Chamaecyparis spp (8-18%mc)	J	Liquidambar styraciflua	A
Chamaecyparis spp (18-28%mc)	C	Lovoa klaineana	J
Chlorophora excelsa	F	Lovoa trichiloides	J
Cordia alliodora	F	Maesopsis eminii	J
Croton megalocarpus	J	Mansonia altissima	B
Cryptomelia japonica	B	Millettia stuhlmannii	A
Cupressus spp	A	Mimusops heckelii	B
Dacrydium franklinii	B	Mitragyna ciliata	E
Dalbergia latifolia	A	Nauclea diderrichii	H
Diospyros virginiana	G	Nesogordonia papaverifera	C
Dipterocarpus (Keruing)	F	Nothofagus cunninghamii	A
Dipterocarpus zeylanicus	A	Ochroma lagopus	A
Distemonanthus benthamianus	C	Ochroma pyramidalis	A
Dracontomelum mangiferum	B	Ocotea rodiaei	C
Dryobalanops spp	A	Ocotea usambarensis	C
Dyera costulata	C	Octomeles sumatrana	E
Entandrophragma angolense	H	Olea hochstetteri	B
Entandrophragma cylindricum	C	Olea welwitschii	C
Entandrophragma utile	J	Palaquium spp	A
Endiandra palmerstoni	C	Paulownia tomentosa	J
Erythrophloeum spp	C	Pericopsis elata	G
Eucalyptus acmenicoides	C	Picea abies	C
Eucalyptus crebra	B	Picea jezoensis (8-18%mc)	J
Eucalyptus diversicolor	A	Picea jezoensis (18-28%mc)	C
Eucalyptus globulus	B	Picea sitchensis	C
Eucalyptus maculata	A	Picaenia excelsa	C
Eucalyptus marginata	C	Pinus caribaea	C
Eucalyptus microcorys	A	Pinus contorta	A
Eucalyptus obliqua	C	Pinus lampertiana	C
Eucalyptus pilularis	C	Pinus nigra	C
Eucalyptus saligna	B	Pinus palustris	C
Eucalyptus wandoo	J	Pinus pinaster	B
Fagus sylvatica	C	Pinus ponderosa	C
Flindersia brayleyana	B	Pinus radiata	C
Fraxinus Americana	B	Pinus spp	B
Fraxinus excelsior	A	Pinus strobus	A
Fraxinus japonicus	A	Pinus sylvestris	A
Fraxinus mardshurica	A	Pinus thunbergii	B
Gonystylus macrophyllum	G	Pipadeniastrum africanum	A
Gossweilodendron balsamiferum	J	Piptadenia africana	A
Gossypiospermum proerox	A	Podocarpus dactyloides	B
Grevillea robusta	C	Podocarpus spicatus	C
Guarea cedrata	H	Podocarpus totara	E
Guarea thomsonii	J	Populus spp	A
Guibortia ehie	B	Prunus avium	J
Hevea brasiliensis	H	Pseudotsuga menziesii	B
Intsia bijuga	B	Pterocarpus angolensis	G

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Pterocarpus indicus	G
Pterocarpus soyauxii	F
Pterygota bequaertii	A
Quercus cerris	E
Quercus delegatensis	C
Quercus gigantean	C
Quercus robur	A
Quercus spp	A
Ricinodendron heudelottii	F
Sarcocephalus diderichii	H
Scottellia coriacea	E
Sequoia sempervirens	B
Shorea smithiana	C
Shorea spp	B
Sterculia rhinopetala	A
Swietenia candollei	A
Swietenia mahogani	B
Syncarpia glomulifera	C
Syncarpia laurifolia	C
Tarrietia utilis	C
Taxus baccata	C
Tectona grandis	F
Terminalia superba	A
Thuja plicata	C
Thujopsis dolabrat	J
Tieghamella heckelii	B
Tilia americana	G
Tilia vulgaris	E
Triplœhiton scleroxylon	G
Tsuga heterophylla	C
Ulmus americana	E
Ulmus procera	E
Ulmus thomasii	E
Xylia dolabriformis	E
Zelkova serrata	B

Std Scale	Species Group								Chip-board
	A	B	C	E	F	G	H	J	
	%H ₂ O								
7	8.2	9.0	8.0	7.1	7.0	11.0	10.5	--	
8	10.0	10.5	9.3	7.5	7.4	11.5	11.0	--	
9	10.8	10.9	9.7	7.9	8.1	12.1	11.6	8.5	
10	11.7	11.5	10.4	8.6	8.8	12.7	12.2	9.4	
11	12.7	12.6	11.3	9.5	9.7	13.4	13.4	10.5	
12	13.6	13.7	12.1	10.5	10.5	14.0	14.3	11.5	
13	14.5	14.5	12.7	11.2	11.2	14.5	15.1	12.5	
14	15.3	15.5	13.4	11.8	11.8	15.0	16.0	13.5	
15	16.3	16.7	14.1	12.5	12.6	15.6	17.0	14.4	
16	16.9	17.5	14.8	13.0	13.2	16.0	17.7	14.9	
17	17.7	18.8	15.7	14.3	13.9	16.6	18.5	15.3	
18	18.2	19.7	16.3	15.0	14.5	17.0	19.1	16.1	
19	19.0	21.0	16.9	15.9	15.2	17.6	20.0	16.7	
20	20.0	22.6	17.8	16.9	16.1	18.4	21.3	17.2	
21	20.8	23.5	18.5	17.6	16.8	19.1	22.3	18.3	
22	21.5	24.5	19.3	18.3	17.4	19.7	23.2	19.1	
23	22.9	26.4	20.2	19.8	18.6	21.2	25.3	19.9	
24	23.5	27.4	20.8	20.4	19.0	22.0	25.8	20.5	
25	24.2	27.8	21.2	21.0	19.4	22.7	26.3	≈23	
26	25.3	29.0	22.4	22.3	20.1	23.9	27.3	--	
27	26.5	--	23.3	23.4	20.8	24.7	28.1	--	
28	28.0	--	24.4	24.8	21.7	25.9	--	--	
29	29.6	--	25.6	26.3	22.9	27.1	--	--	

NOTES

- Protimeter Mini, Surveymaster and MMS instruments display Standard Scale A. Protimeter Timbermaster displays all scales.
- The calibration data in this table are based on standard tests by oven-drying of commercial samples of the various wood species, between 7% and fibre saturation. Above fibre saturation point (25-30%) readings are approximate only and generally apply to wood that has dried and been re-wetted.
- The instrument is calibrated for wood at 20°C (68°F). If the temperature of the wood varies by more than 5°C, the meter reading can be corrected approximately by adding 1/2% for every 5°C below 20°C or subtracting 1/2% for every 5°C above 20°C.
- Readings higher by 1-2% may be obtained where wood has been impregnated with a water-borne preservative.
- High readings obtained with some ply-woods of peculiar composition must be treated with caution.

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4 RF Meter Description

4.1 Features

- 1 The 'RF' moisture meter is an EVO SM [Saw Mill] model (or its predecessor, the EVO25 model), manufactured by Merlin Technology GmbH, Austria. It is powered by 4 x AAA 1.5v alkaline batteries.



- 2 This model of moisture meter measures capacitance, using a high frequency dielectric field. The dielectric constant of the material is then related (via the internal software and taking into account species density and material temperature) to a moisture level.
- 3 The meter dimensions are 124 x 71 x 30 mm and the sensor pad on the base of the meter measures 66 x 103 mm.



- 4 The meter can accurately measure between 2%mc and the fibre saturation point of the wood. The entire measuring range of the meter is 2%mc to 99%mc. However (as is the case with resistance moisture meters) any reading over the fibre saturation point of the wood cannot be deemed to be accurate.
- 5 The maximum read depth of the meter is 40mm.
- 6 The readings obtained are near instantaneous. Consequently, it is possible to move the meter across the surface of the wood to identify wet pockets. However, with poles, it is advisable to opt for multiple separate readings.

4.2 Operating Parameters

- 1 The meter can be used to obtain accurate measurements only within the temperature range 0°C and 40°C. It can be stored at temperatures between -20°C and +60°C (with the batteries removed).

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- 2 The meter is pre-programmed to measure across a wood density range of 175 - 1075 kg/m³ (0.10 - 0.62 oz. /in³).
- 3 The minimum thickness of wood that can be accurately measured is 25mm.

5 Using an RF Meter on Poles

5.1 Device Limitations

- 1 **LIMITATIONS:** The accuracy of this type of meter can be adversely affected by the following factors:
 - If any part of the sole plate is not directly contacting the wood surface then the void space between the sole plate and the wood will be read as dry wood. So, for example on a 6 metre light pole, very little of the sole plate will be in contact with the wood and the resulting reading will be much drier than the actual moisture content of the wood.
 - If the wood that is being measured has any surface moisture present then this will deliver an artificially wet reading – because the meter reads everything in the footprint of the sole plate and within the distance of 0mm to 40mm.
- 2 Because of the limitations described, there is little point in calibrating this type of meter with the approved check box for use on poles because the check box delivers a reading that is obtained from a completely dry material that is in contact with the full face of the sole plate.
- 3 Instead, an accurate RF meter reading is achieved via correspondence with the reading delivered by the Pin Meter.

5.2 Achieving Meter Correspondence

- 1 Meter correspondence must be conducted upon each of the following:
 - At shift commencement.
 - Following a change in the class of pole being measured.
 - Following a change in the batch of poles (for example a change in the source pole boat)
 - Where the RF meter is dropped (or otherwise suffers an impact)

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2 Meter correspondence is achieved as follows:



- Inspect and calibrate the Pin Meter as described in section 3.



- Inspect the RF meter for signs of damage, contamination or moisture. Pay particular attention to the sensing pad, ensuring that it is in good condition and free from dirt and contaminants (such as resin).



- Hold the RF meter in such a position that nothing is in the vicinity of the sensor pad, for example hold the meter with the sensor pad pointing away from you and in the air.



- Switch the meter on by pressing the  power/ light button. Keep the button pressed until the LCD display illuminates (approx. 3 seconds). The meter will now run through a set-up procedure, during which time the serial number, software version and battery status are displayed.



- When the meter is ready for operation, the screen below will be displayed.



- Drive the pins of the Pin Meter into a surface-dry pole until they are fully inserted into the wood and note the moisture reading.



- Place the RF meter on the same spot on the wood where you applied the Pin Meter and apply light pressure (approx. 4kg). The moisture content of the wood will be displayed.



- If the moisture reading of the RF meter is different from the reading of the Pin Meter then press the up  or down  button to scroll through the different species until the two readings are the same.

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- Repeat this comparison on at least 3 more poles and until you are comfortable that correspondence has been achieved.
- Normal readings are displayed in solid characters. However, if an obtained reading is above the fibre saturation of the wood, the characters in the display will become faded and grey and the characters will blink repeatedly.

5.3 Selection of Testing Locations

- 1 The meter calculates moisture content taking into account the density of the wood. For this reason you must avoid the following locations when obtaining measurements:
 - Surface checks (also known as weathering checks). The meter will use the density of the void space in its calculations and return a potentially inaccurate reading.
 - Knots – The varying densities within the knot (heartwood, sapwood and deviated grain) will affect the calculations and return a potentially inaccurate reading.
 - Uneven surfaces – Accurate measurement requires that as much of the sensor pad as possible is in contact with the wood. If you try to measure across a raised feature in the wood or over an uneven area, the meter will use the density of the void space between the wood and the sensor plate in its calculations and will return a potentially inaccurate reading.
- 2 The ideal conditions for obtaining accurate measurement are:
 - Flatter areas that allow more of the sensing pad to be in contact with the wood.
 - Areas that are clear of surface checks, knots, in-barks or felling machine indentations.
 - Level areas, not straddling different heights (for example, if pole dressing has resulted in a spiral of slightly raised surface, care should be taken to avoid placing the sensing plate across the two heights, as a void will be created under the sensing plate).
 - Wood that is not surface wet (for example a pole that has just been in the rain would have heavy surface wetness and this would be reflected in a very high reading).

6 Using an RF Meter on Flat Timbers

6.1 Pre-Use Verification of Calibration

- 1 At the beginning of each day, you must check that the calibration of the meter is within the required parameters, using the dedicated calibration check box that is provided for this purpose.

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Effective Date: Mar 2020		Revision Date: Nov 2023	

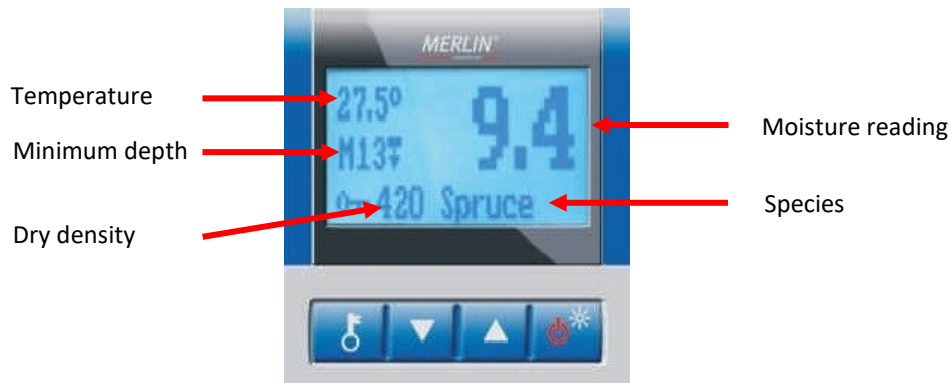
2 You must conduct the calibration check as follows:

- DO NOT place the test board on a metal surface during testing, as this can affect the meter readings.
- Remove all dirt and foreign matter from the sensor pad on the meter and also from the test board.
- Switch on the meter (whilst out of the box).
- When the display stabilises, select 'Pine' as the wood species (dry density 460).
- Place the meter into the calibration check box (with the entire sensor pad touching the moisture-stable testing element).
- Apply light pressure.
- Note the moisture reading displayed.
- If the moisture reading is 14.0% (or within a tolerance of 1.3%) then you can use the meter. If this is not the case, you must not use the meter and you must be return it to the Production Supervisor, together with a description of the defect.

6.2 Operating Procedure

- 1 Inspect the meter for signs of damage, contamination or moisture. Pay particular attention to the sensing pad, ensuring that it is in good condition and free from dirt and contaminants (such as resin).
- 2 Hold the meter in such a position that nothing is in the vicinity of the sensor pad, for example hold the meter with the sensor pad pointing away from you and in the air.
- 3 Switch the meter on by pressing the  power/ light button. Keep the button pressed until the LCD display illuminates (approx. 3 seconds). The meter will now run through a self-calibration procedure, during which time the serial number, software version and battery status are displayed.
- 4 When the meter is ready for operation, the screen below will be displayed.

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- 5 To change the wood species, press the up  or down  button.
- 6 If you need to take a measurement in a location that is out of your line of sight, you can place the meter in position for the reading and then press the hold  button (once) to freeze the measurement on the screen.
- 7 If you need to light the display, press the  power/ light button briefly.
- 8 Place the meter on the wood to be measured and apply light pressure (approx. 4kg). The moisture content of the wood is immediately displayed.
- 9 To obtain further readings just move the meter and apply light pressure again.
- 10 Normal readings are displayed in solid characters. However, if an obtained reading is above the fibre saturation of the wood, the characters in the display will become faded and grey and the characters will blink repeatedly.
- 11 If the meter is dropped (or otherwise suffers an impact) you must be re-check the calibration using the approved calibration check box.

6.3 Selection of Testing Locations

- 1 The meter calculates moisture content taking into account the density of the wood. For this reason you must avoid the following locations when obtaining measurements:
 - Surface checks (also known as weathering checks). The meter will use the density of the void space in its calculations and return a potentially inaccurate reading.
 - Knots – The varying densities within the knot (heartwood, sapwood and deviated grain) will affect the calculations and return a potentially inaccurate reading.
 - Uneven surfaces – Accurate measurement requires that as much of the sensor pad as possible is in contact with the wood. If you try to measure across a raised feature in the wood or over an uneven area, the meter will use the density of the void space between

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the wood and the sensor plate in its calculations and will return a potentially inaccurate reading.

- Small diameter round timbers – because the sensor plate is flat and the timber surface is sharply curved, it is not possible to keep a sufficient amount of the sensor in contact with the timber and the meter will use the density of the void space in its calculations and return a potentially inaccurate reading.

2 The ideal conditions for obtaining accurate measurement are:

- Flatter areas that allow more of the sensing pad to be in contact with the wood.
- Areas that are clear of surface checks, knots, in-barks or felling machine indentations.
- Level areas, not straddling different heights (for example, if pole dressing has resulted in a spiral of slightly raised surface, care should be taken to avoid placing the sensing plate across the two heights, as a void will be created under the sensing plate).
- Wood that is not surface wet (for example a pole that has just been in the rain would have heavy surface wetness and this would be reflected in a very high reading).

7 Oven Testing Procedure

7.1 Obtaining Samples

- 1 Samples of 75mm, or the full depth of the sapwood, whichever is greater, are taken from the poles, at a point not less than 1.5m from the butt of the pole. This will be achieved by means of a test borer, consisting of a hollow auger and extractor. The poles are then plugged – preferably with a preservative impregnated plug.

7.2 Method

- 1 When the sample is taken it is placed into a plastic bag and sealed. (The heat from the boring allows water to evaporate from the sample).
- 2 Immediately after the samples are taken, they are weighed (*m1*). If more than one sample is being tested, they are numbered for re-weighing, after drying.
- 3 The samples are dried in an oven, at a temperature of 103°C +/- 2°C, until the mass is constant.
- 4 The samples are weighed immediately after removal from the oven (*m2*).

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7.3 Calculation

- The moisture content (**w**) is expressed as a percentage of the oven dry mass (**m2**) using the following formula:

$$w \text{ moisture content} = \frac{m1 \text{ (sample)} - m2 \text{ (oven dried sample)}}{m2 \text{ (oven dried sample)}} \times 100$$

Documents Replaced by this Module

Document Ref.	Document Title
QTG91	Use of RF Moisture Meters
QTG60	Preservation - Moisture Content of Poles

Updates

Date	Amendment	By
1-3-20	Original Document	DL
13-7-21	Details of testing locations added at section 2.	DL
24-11-23	Principle of correspondence introduced for RF checking of poles	DL

Next Review 2025 – unless otherwise required